

Aditya Agarwal

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Summary

Computer Science student with strong foundations in Data Structures and Algorithms, Object-Oriented Programming, and System Design (High-Level & Low-Level Design). Shipped reusable Python modules used by internal teams, optimized an ETL pipeline by 73%, and designed a scalable system handling 10M+ daily writes. Seeking Software Development Engineer (SDE) Intern role for Summer 2026.

Technical Skills

Languages: Python, C++, C, SQL
Core CS: Data Structures & Algorithms, Object-Oriented Programming, System Design (High-Level & Low-Level Design)
Backend / Systems: Redis, SQLite, Consistent Hashing, REST APIs, ETL Pipelines
AI / ML (Applied): Scikit-learn, Pandas, OpenAI API, LangChain
Tools: Git, GitHub, Postman, Jupyter

Professional Experience

Python Developer Intern Oct 2024 – Nov 2024

CodSoft

- Engineered 5 reusable Python modules for data validation and file I/O, reducing runtime errors by 35% across 3 internal projects.
- Optimized legacy ETL script from 45s to 12s (73% improvement) using batch processing and caching.

Co-Founder & Technical Lead Feb 2025 – Present

PANKH

- Built and shipped an internal order management system (SQLite + Python CLI) handling 150+ orders with 99% data accuracy.
- Led technical decisions end-to-end as Technical Lead for a 2-person engineering team.

Projects

URL Shortener – System Design Redis, Base62, Consistent Hashing

- Designed scalable architecture (similar to bit.ly) for 10M+ daily writes with under 10ms redirect latency and 99.9% uptime via Redis cache and replication.

Fraction Arithmetic Library (OOP) C++

- Built C++ class library with overloaded operators for exact fraction arithmetic, following SOLID principles; covered by 60+ unit tests.

Customer Churn Prediction Python, Scikit-learn, Pandas

- Built and deployed a Random Forest classifier achieving 86% accuracy and 0.81 F1-score on a 10,000-row dataset.

Education

B.Tech – Computer Science Engineering (AI & ML) 2024 – 2028 (Expected)

Dr. A.P.J. Abdul Kalam Technical University

CGPA: **8.23 / 10**

Senior Secondary (Class XII) – Science 2023 – 2024

Wendy High School (ISC)

87%

Certifications & Achievements

Smart India Hackathon 2025 – Finalist
Agile Project Management – HP LIFE
Python for AI & ML – Be10x
JEE Mains 2025 Qualified – Top 18 Percentile

Additional

Languages: English, Hindi
Interests: Competitive programming (3-star CodeChef), System design case studies, Open source